Tutorial 1

AMA3707 Real Analysis

1 Sets and Functions

Exercise 1.1. [exc:abbott-1-2-3] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.2.3.

(a) If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$ and every A_n is infinite, then $\bigcap_{n=1}^{\infty} A_n$ is infinite.
Given that $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ are all containing infinite number of elements.
This is a nested sequence that

$$\bigcap_{n=1}^{m} A_n = A_m$$

Notice that for all sets A_n, it contains infinite number of elements. Meaning that $m \to \infty$, A_m will still contain infinite number of elements.
Also,

$$\bigcap_{n=1}^{\infty} A_n = \lim_{m\to\infty} \bigcap_{n=1}^{m} A_n = A_m$$

So,

$$\left| \bigcap_{n=1}^{\infty} A_n \right| = |A_m| = \infty$$

Statement (a) is true.

(b) If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$ and every A_n is finite and nonempty, then $\bigcap_{n=1}^{\infty} A_n$ is finite and nonempty.
By the same argument,

$$\bigcap_{n=1}^{\infty} A_n = \lim_{m\to\infty} \bigcap_{n=1}^{m} A_n = A_m, \quad 0 < |A_m| < \infty$$

So,

$$\bigcap_{n=1}^{\infty} A_n = A_m \quad \text{is finite and nonempty.}$$

Statement (b) is true.

(c) $A \cap (B \cup C) = (A \cap B) \cup C$.
By membership under classical logic, every element x in the chosen universe satisfies exactly one of the following eight conditions:

1. $x \in A$ and $x \in B$ and $x \in C$
2. $x \in A$ and $x \in B$ and $x \notin C$
3. $x \in A$ and $x \notin B$ and $x \in C$
4. $x \in A$ and $x \notin B$ and $x \notin C$
5. $x \notin A$ and $x \in B$ and $x \in C$
6. $x \notin A$ and $x \in B$ and $x \notin C$
7. $x \notin A$ and $x \notin B$ and $x \in C$
8. $x \notin A$ and $x \notin B$ and $x \notin C$

Define

$$D = A \cap (B \cup C), \quad E = (A \cap B) \cup C$$

Denote the above 8 conditions as $P_1, P_2, \ldots, P_8$.
By using definition of definition intersection and definition union, we can see that

- $\forall x \in P_1 \implies x \in D$ and $x \in E$
- $\forall x \in P_2 \implies x \in D$ and $x \in E$
- $\forall x \in P_3 \implies x \in D$ and $x \in E$
- $\forall x \in P_4 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_5 \implies x \notin D$ and $x \in E$
- $\forall x \in P_6 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_7 \implies x \notin D$ and $x \in E$
- $\forall x \in P_8 \implies x \notin D$ and $x \notin E$

So, $D = P_1 \cup P_2 \cup P_3$ and $E = P_1 \cup P_2 \cup P_3 \cup P_5 \cup P_7$.
$\exists x, x \in E \wedge x \notin D$ such as $x \in P_5$ or $x \in P_7$, meaning that $D \subset E$.
According to the definition of set equality, $D \neq E$.
Statement (c) is false.

(d) $A \cap (B \cap C) = (A \cap B) \cap C$.
By membership under classical logic, every element x in the chosen universe satisfies exactly one of the following eight conditions:

1. $x \in A$ and $x \in B$ and $x \in C$
2. $x \in A$ and $x \in B$ and $x \notin C$
3. $x \in A$ and $x \notin B$ and $x \in C$
4. $x \in A$ and $x \notin B$ and $x \notin C$
5. $x \notin A$ and $x \in B$ and $x \in C$
6. $x \notin A$ and $x \in B$ and $x \notin C$
7. $x \notin A$ and $x \notin B$ and $x \in C$
8. $x \notin A$ and $x \notin B$ and $x \notin C$

Define

$$D = A \cap (B \cap C), \quad E = (A \cap B) \cap C$$

Denote the above 8 conditions as $P_1, P_2, \ldots, P_8$.
By using definition of definition intersection and definition union, we can see that

- $\forall x \in P_1 \implies x \in D$ and $x \in E$
- $\forall x \in P_2 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_3 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_4 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_5 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_6 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_7 \implies x \notin D$ and $x \notin E$
- $\forall x \in P_8 \implies x \notin D$ and $x \notin E$

So, $\forall x \in D \implies x \in E$, ie $D \subseteq E$. Also, $\forall x \in E \implies x \in D$, ie $E \subseteq D$.
According to the definition of set equality, $D = E$.
Statement (d) is true.

(e) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
By membership under classical logic, every element x in the chosen universe satisfies exactly one of the following eight conditions:

1. $x \in A$ and $x \in B$ and $x \in C$
2. $x \in A$ and $x \in B$ and $x \notin C$
3. $x \in A$ and $x \notin B$ and $x \in C$
4. $x \in A$ and $x \notin B$ and $x \notin C$
5. $x \notin A$ and $x \in B$ and $x \in C$
6. $x \notin A$ and $x \in B$ and $x \notin C$
7. $x \notin A$ and $x \notin B$ and $x \in C$
8. $x \notin A$ and $x \notin B$ and $x \notin C$

Exercise 1.2. [exc:abbott-1-2-7] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.2.7.

For a function f with domain D and a set $A \subseteq D$, use

$$f(A) = \{f(x) : x \in A\}.$$

(a)

$$f(x) = x^2, \qquad A = [0, 2], \qquad B = [1, 4].$$

Find $f(A)$ and $f(B)$, and determine whether

$$f(A \cap B) = f(A) \cap f(B) \quad \text{and} \quad f(A \cup B) = f(A) \cup f(B).$$

Since f is a one to one map $[0, 4]$
Where, $A, B \in [0, 4]$.
We can find the image of A and B by evaluating the function at the endpoints of the intervals. Trivially:

$$f(A) = [0, 4], \quad f(B) = [1, 16]$$

Where $A \cap B = [1, 2]$, so $f(A \cap B) = [1, 4]$.
We can also find $f(A) \cap f(B) = [0, 4] \cap [1, 16] = [1, 4]$.
$\boxed{\text{So, } f(A \cap B) = f(A) \cap f(B).}$
Where $A \cup B = [0, 4]$, so $f(A \cup B) = [0, 16]$.
We can also find $f(A) \cup f(B) = [0, 4] \cup [1, 16] = [0, 16]$.
$\boxed{\text{So, } f(A \cup B) = f(A) \cup f(B).}$
According to proposition bijection preimage image,

(b) Find A and B such that

$$f(A \cap B) \neq f(A) \cap f(B).$$

For $f = x^2$, we can have some part which is not one-to-one.

$$\begin{aligned} A &= [0, 1], & B &= [-1, 0], \\ A \cap B &= \{0\}, & f(A \cap B) &= f(\{0\}) = \{0\}, \\ f(A) \cap f(B) &= f([0, 1]) \cap f([-1, 0]) = [0, 1] \cap [0, 1] = [0, 1], \\ [0, 1] &\neq \{0\}. \end{aligned}$$

(c) For an arbitrary function $g : \mathbb{R} \to \mathbb{R}$ and all $A, B \subseteq \mathbb{R}$, prove that

$$g(A \cap B) \subseteq g(A) \cap g(B).$$

Proof. According to definition of image,

$$g(A) = y : g(x), x \in A \quad g(B) = y : g(x), x \in B \quad g(A \cap B) = y : g(x), x \in A \cap B$$

Assume

$$g(A \cap B) \supset g(A) \cap g(B) \implies \exists y, y \notin g(A) \cap g(B) \wedge y \in g(A \cap B)$$

Then, $y \in g(A \cap B)$ means that $\exists x, x \in A \cap B$ and $g(x) = y$. Since $x \in A \cap B$, we have $x \in A$ and $x \in B$. Therefore, $g(x) \in g(A)$ and $g(x) \in g(B)$, which implies $y \in g(A) \cap g(B)$. This is a contradiction.
Therefore, $g(A \cap B) \subseteq g(A) \cap g(B)$. $\square$

(d) Determine and prove the relationship between

$$g(A \cup B) \quad \text{and} \quad g(A) \cup g(B)$$

for an arbitrary function g.

$$\forall y \in g(A) \cup g(B), \exists x \in A \cup B, g(x) = y \implies y \in g(A \cup B) \implies g(A \cup B) \supseteq g(A) \cup g(B)$$

Exercise 1.3. [exc:abbott-1-2-9] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.2.9.

For a function $f : D \to \mathbb{R}$ and a set $B \subseteq \mathbb{R}$, use

$$f^{-1}(B) = \{x \in D : f(x) \in B\}.$$

(a)

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2, \qquad A = [0, 4], \qquad B = [-1, 1].$$

Find $f^{-1}(A)$ and $f^{-1}(B)$, and determine whether

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B) \quad \text{and} \quad f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B).$$

(b) For an arbitrary function $g : \mathbb{R} \to \mathbb{R}$ and all $A, B \subseteq \mathbb{R}$, prove that

$$g^{-1}(A \cap B) = g^{-1}(A) \cap g^{-1}(B) \quad \text{and} \quad g^{-1}(A \cup B) = g^{-1}(A) \cup g^{-1}(B).$$

2 Suprema and Infima

Exercise 2.1. [exc:abbott-1-3-3] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.3.3.

(a) Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let

$$B = \{b \in \mathbb{R} : b \leq a \text{ for every } a \in A\}.$$

Prove that

$$\sup B = \inf A.$$

Proof. According to the definition of bounded below, there exists a lower bound m for A st

$$\forall x \in A, m \leq x$$

According to the definition of set B, we have $m \in B$, or equiv, B is the set of all lower bounds of A.
Which means the biggest item in B is the infimum of A.
By the same argument, $\inf A$ is the supremum of B.
So, $\sup B = \inf A$. □

(b) Using part (a), explain why the Completeness Axiom only needs to assert the existence of suprema; the corresponding existence of infima follows.
Since $\sup B = \inf A$, if we have the existence of suprema, we can find the infima by finding the supremum of the set of lower bounds.

Exercise 2.2. [exc:abbott-1-3-4] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.3.4.

Let $A_1, A_2, \ldots$ be nonempty subsets of $\mathbb{R}$, each bounded above.
(a) Determine

$$\sup(A_1 \cup A_2) \quad \text{and} \quad \sup\left(\bigcup_{k=1}^{n} A_k\right).$$

$$\sup(A_1 \cup A_2) = \max\{\sup A_1, \sup A_2\}$$
$$\sup\left(\bigcup_{k=1}^{n} A_k\right) = \sup(\{\sup A_1, \sup A_2, \ldots, \sup A_n\})$$

(b) Determine whether the finite-union formula extends to

$$\sup\left(\bigcup_{k=1}^{\infty} A_k\right).$$

$$\sup\left(\bigcup_{k=1}^{\infty} A_k\right) = \sup(\{\sup A_1, \sup A_2, \ldots, \sup A_n, \ldots\})$$

Exercise 2.3. [exc:abbott-1-3-11] *Source:* Stephen Abbott, *Understanding Analysis*, Second Edition, Exercise 1.3.11.

Decide whether each statement is true or false, and justify the answer.
(a) If A and B are nonempty and bounded, and $A \subseteq B$, then

$$\sup A \leq \sup B.$$

According to the definition of upper bound, the set of upper bound of A and B is

$$U_A = \{x \in \mathbb{R} : x \geq a, \forall a \in A\}, \quad U_B = \{x \in \mathbb{R} : x \geq b, \forall b \in B\}$$

Since $A \subseteq B$, every element of A is also an element of B. Therefore, any upper bound of B is also an upper bound of A. This means that $U_B \subseteq U_A$, and thus $\sup A \leq \sup B$.
Statement (a) is true.

(b) If $\sup A < \inf B$, then

$$\exists c \in \mathbb{R}, \quad \forall a \in A, \ \forall b \in B, \qquad a < c < b.$$

Notice that $\sup A$ is one of the upper bounds of A, meaning that:

$$\forall x \in A, x < \sup A$$

Also, for $\inf B$, it is one of the lower bounds of B, meaning that:

$$\forall x \in B, x > \inf B$$

Notice that $\sup A < \inf B$, we can choose $c = \frac{\sup A + \inf B}{2}$, which is a number between $\sup A$ and $\inf B$.
Then, for all $a \in A$, we have $a < \sup A < c$, and for all $b \in B$, we have $c < \inf B < b$.
Statement (b) is true.

(c) If

$$\exists c \in \mathbb{R}, \quad \forall a \in A, \ \forall b \in B, \qquad a < c < b,$$

then

$$\sup A < \inf B.$$